

# ROMIR MALIK

Utrecht, Netherlands | +31 6 34386359 | [romir.ds10@gmail.com](mailto:romir.ds10@gmail.com) | [linkedin.com/in/romirmalik](https://www.linkedin.com/in/romirmalik) | [github.com/kruuusher13](https://github.com/kruuusher13) | [kruuusher13.github.io/Portfolio](https://kruuusher13.github.io/Portfolio)

MSc Applied Data Science, Utrecht University, 2026. I measure what computation costs, in energy, latency and euros, and design against it: FPGA inference at ISRO, pipeline energy on the Dutch national supercomputer at TNO, and the measurement layer behind a product in production.

**Work authorisation:** recent Dutch MSc graduate, eligible for the orientation year and for kennismigrant sponsorship.

## PUBLICATIONS

**Complexity-aware energy estimation of the data curation pipeline for GPT-NL on HPC clusters.** TNO and Utrecht University. MSc thesis; code, CLI and model weights [1]. **In preparation**

**Rope-augmented path following and control of a remotely operated underwater vehicle using vision for stilling basin surveillance.** R. Malik et al., IEEE APSCON 2024 [2]. **Jan 2024**

## PROFESSIONAL EXPERIENCE : HPC to silicon

**TNO**, Netherlands **Apr 2026 to Jul 2026**  
MSc Thesis Researcher

- Priced a full GPT-NL data curation run on Snellius in joules, euros and CO2 before launch. A per stage law  $E = c_0 + c_1 \cdot n$  over five stages predicts a **held out 400k document run to 2.8% error** from calibration runs of up to 100k [1].
- Built an **extended Kalman filter with an innovation gate** as the live mechanism, blending the prior with EAR per job telemetry and rejecting implausible readings: final error moved from +93% to +8% on a contaminated run [1], [3].
- Ran it **across four corpora differing 100 fold in mean document length**, so estimates transfer per character, not per document. Shipped as a command line tool over six estimators, the physics based linear model beating all five learned ones [1].

**ConcertLab**, Netherlands **Sep 2026 to present**  
Data Scientist

- Work **directly with the CEO on data informed strategy**: built the four year P&L, scenario and unit economics model behind a live activation round, running capacity and breakeven across two scenarios [4].
- Built the measurement layer for a studio with no instrumentation: **11 containerised services and 12 always on daemons unifying 5 enquiry channels** on Postgres and Redis, making channel attribution observable for the first time.

**Torxflow**, Netherlands **May 2026 to present**  
Founding Engineer

- Founding engineer on a **garage management platform built around three intelligence modules**: smart scheduling, vehicle intelligence and AI diagnosis, over the systems a Dutch workshop already uses [5].
- Vehicle intelligence resolves a licence plate to service history, profile and parts availability, and scheduling writes into the existing DMS. Telephone intake runs lookup, parts search and labour time pricing concurrently at **390 ms median response and 94% intent confidence**.
- **Graduated from the UtrechtInc validation programme**, Utrecht's startup incubator.

**Council of Scientific and Industrial Research (CSIR), National Institute of Oceanography**, Goa, India **Sep 2023 to May 2024**  
Machine Learning and Robotics Intern

- Built perception and avoidance for an unmanned underwater vehicle with no GPS, on **monocular depth rather than stereo** because a second pressure housing adds a failure mode: ZoeDepth and MiDaS in a GStreamer and ROS pipeline, sparse structure, potential fields for avoidance [6].
- Chose potential fields for graceful degradation: a lost detection bends the path rather than breaking it. **Over 90% detection in turbid water and about 40% lower planning latency**, across 50 scenarios in Gazebo [7], [8].

**Indian Space Research Organisation (ISRO), Space Applications Centre**, Ahmedabad, India **May 2023 to Aug 2023**  
Research Intern

- Moved deep learning object detection onto the FPGA fabric of a Xilinx Zynq MPSoC so a satellite sees without a downlink: under 15 ms end to end at 60+ FPS, a 4x speed up over CPU at roughly 3x performance per watt.
- Identified **the data path outside the offloaded region, not the kernel, as the Amdahl bound**: moving feature maps across AXI by DMA with double buffering hit the target with the kernel untouched. Feasibility report adopted by senior SAC scientists.

## EDUCATION

**Utrecht University**, MSc Applied Data Science **Sep 2025 to Aug 2026**

Thesis at TNO. Built **Chess-Bot-20M**, which reframed move selection as classification over the 1,968 legal moves rather than as generation and **won the INFOMTALC 2026 tournament at 38.9M parameters, 180x smaller than the 7B baselines** it beat [9].

**Manipal University Jaipur**, India, BTech in Data Science and Engineering **2020 to 2024**

## LEADERSHIP AND ENTREPRENEURSHIP

**Utrecht Data Science Community**, Head of External Relations **Sep 2025 to present**

Own the industry side of Utrecht's data science community: **partnerships with more than 10 organisations and a pipeline connecting more than 500 students** to corporate recruiters, with event attendance up 40% year on year.

**UtrechtInc**, Validation Programme with Torxflow, graduated. Utrecht's startup incubator. **2026**

## TECHNICAL SKILLS

**Tools:** Python, PyTorch, SQL, C++, SLURM, Snellius, Kalman filtering, ROS 2, Zynq MPSoC, Vivado, transformers, scikit-learn, ORB-SLAM3, OpenCV, Docker, Postgres, Redis, Verilog, CUDA C. **Spoken:** Dutch (beginner, taking classes), English (fluent), Hindi (native).